

Complexity Theory

Quiz 1 Set A

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1 **mark** for the right answer and **-(1/3)** for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name:

Roll No:

Duration: 40 Minutes

Marks:

1. $NP \neq NP\text{-complete}$.

True. This was tricky. **NP** contains language Σ^* , that is, the decision problem in which for every input the answer is "Yes". Now Σ^* can never be **NP-complete** as any non-trivial language in **NP** with "Yes" and "No" instances cannot be reduced to Σ^* as Σ^* has no "No" instances.

2. For any two languages L_1 and L_2 , if $L_1 \leq L_2$, then $\overline{L_2} \leq \overline{L_1}$.

False. $\overline{L_2} \leq \overline{L_1}$ means $L_2 \leq L_1$. The statement is basically saying that for any two languages L_1 and L_2 , if L_1 is reducible to L_2 then L_2 is reducible to L_1 . The statement is clearly false. Let L_1 be some decidable language and L_2 be HALT. Then L_1 can be reduced to L_2 , but not vice versa.

3. If $P = NP$, then $\overline{SAT} \in P$.

True. If $P = NP$, then SAT will belong to **P**. It is easy to see that if any language is in **P**, then its complement is also in **P**. Just reverse the output of TM.

4. The below claim will not be true, if T is not a time-constructible function.

Claim: For any $f: \{0,1\}^* \rightarrow \{0,1\}^*$ and time-constructible function $T: \mathbb{N} \rightarrow \mathbb{N}$, if f is computable in time $T(n)$ by a TM M using k tapes, then it is computable in time $O(k \cdot T(n)^2)$ by using a **single-tape** TM M' .

False. The claim does not require T to be time-constructible. We discussed this in the class.

5. Let $Bipartite = \{ \langle G \rangle \mid G \text{ is an undirected bipartite graph.} \}$. Then $Bipartite \notin NP$.

False. $Bipartite$ is in **P**, hence, also in **NP**.

6. If $\mathbf{NP} = \mathbf{EXP}$, then $\mathbf{SAT}$ does not have a polynomial time algorithm.

True. Because $\mathbf{SAT}$ in $\mathbf{P}$ will imply, $\mathbf{P} = \mathbf{NP} = \mathbf{EXP}$. But we know that $\mathbf{P} \subset \mathbf{EXP}$.

7. If L can be decided by a TM M with 10 tapes that runs in $T(n)$ time, where $T(n)$ is time-constructible, then there exists an oblivious TM M' with only two tapes that can decide L and runs in $O(T(n)^4)$ time.

True. Multiple tape to single tape requires $O(T(n)^2)$ time. Single tape to single tape oblivious require $O(T(n)^4)$ time.

8. $GI \in \mathbf{NP-complete}$, where $GI = \{\langle G_1, G_2 \rangle \mid \text{Graphs } G_1 \text{ and } G_2 \text{ are isomorphic}\}$.

Unknown. Was discussed in the class.

9. $\mathbf{PATH} \notin \mathbf{NP-complete}$, where $\mathbf{PATH} = \{\langle G, u, v \rangle \mid G \text{ is an undirected graph and } u \text{ and } v \text{ are two of its vertices such that there exists a path from } u \text{ to } v\}$.

Unknown. First observe that, if $\mathbf{P} = \mathbf{NP}$, then all the problems in $\mathbf{P}$ except Σ^* and $\emptyset$ are **NP-complete**. This is because in order to reduce some problem in $\mathbf{NP}$ ($= \mathbf{P}$) to any problem in $\mathbf{P}$ one can first simply solve it in polytime and then map it to appropriate instance. If $\mathbf{PATH} \notin \mathbf{NP-complete}$ is true, then $\mathbf{P} \neq \mathbf{NP}$ as $\mathbf{P} = \mathbf{NP}$ will contradict $\mathbf{PATH} \notin \mathbf{NP-complete}$. On the other hand, $\mathbf{PATH} \notin \mathbf{NP-complete}$ is false, that is, $\mathbf{PATH} \in \mathbf{NP-complete}$, then $\mathbf{P} = \mathbf{NP}$. Hence, in both the cases, whether $\mathbf{PATH} \notin \mathbf{NP-complete}$ is true or false, it will settle the $\mathbf{P}$ vs $\mathbf{NP}$ problem.

10. There cannot exist two **NP-complete** languages L_1 and L_2 , such that $L_1 \in \mathbf{P}$, but $L_2 \notin \mathbf{P}$.

True. If L_1 is in $\mathbf{P}$, then L_2 must be in $\mathbf{P}$ and vice versa as L_1 and L_2 being **NP-complete** are reducible to each other in polynomial time.

Complexity Theory

Quiz 1 Set B

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1 **mark** for the right answer and **-(1/3)** for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name:

Duration: 40 Minutes

Roll No:

Marks:

1. If $NP = EXP$, then SAT does not have a polynomial time algorithm.

True.

2. $NP = NP\text{-complete}$.

False.

3. If $P = NP$, then $\overline{SAT} \in P$.

True.

4. $PATH \notin NP\text{-complete}$, where $PATH = \{ \langle G, u, v \rangle \mid G \text{ is an undirected graph and } u \text{ and } v \text{ are two of its vertices such that there exists a path from } u \text{ to } v \}$.

Unknown.

5. Let $Bipartite = \{ \langle G \rangle \mid G \text{ is an undirected bipartite graph.} \}$. Then $Bipartite \in NP$.

True.

6. For any two languages L_1 and L_2 , if $L_1 \leq L_2$, then $\overline{L_2} \leq \overline{L_1}$.

False.

7. If L can be decided by a TM M with 10 tapes that runs in $T(n)$ time, where $T(n)$ is time-constructible, then there exists an oblivious TM M' with only two tapes that can decide L and runs in $O(T(n)^4)$ time.

True.

8. There exists two **NP-complete** languages L_1 and L_2 , such that $L_1 \in \mathbf{P}$, but $L_2 \notin \mathbf{P}$.

False.

9. The below claim will not be true, if T is not a time-constructible function.

Claim: For any $f: \{0,1\}^* \rightarrow \{0,1\}^*$ and time-constructible function $T: \mathbb{N} \rightarrow \mathbb{N}$, if f is computable in time $T(n)$ by a TM M using k tapes, then it is computable in time $O(k \cdot T(n)^2)$ by using a **single-tape** TM M' .

False.

10. $GI \in \mathbf{NP-complete}$, where $GI = \{\langle G_1, G_2 \rangle \mid \text{Graphs } G_1 \text{ and } G_2 \text{ are isomorphic}\}$.

Unknown.